

K2-29b

R-2054

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_230120 · created 2026-08-09T00:38:39+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459964.7892 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.0076
Transit depth (Rp/Rs)^2	1.56 +/- 0.19 [%]
Orbital Inclination (inc)	87.9 +/- 0.84
Airmass coefficient 1 (a1)	1.0281 +/- 0.0076
Airmass coefficient 2 (a2)	-0.0238 +/- 0.005
Scatter in the residuals of the lightcurve fit is	0.74 %
Best Comparison Star	#1 - [576, 163]
Optimal Aperture	3.7
Optimal Annulus	5.11
Transit Duration (day)	0.0986 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.0076 vs 0.1419 (deviation 2.11σ)
Duration fit vs published	0.0986 d vs 0.0925 d (deviation 1.14σ)
Mid-time O–C	-6.8 min
Depth SNR	15.6
Residual scatter	0.74 %

Light curve

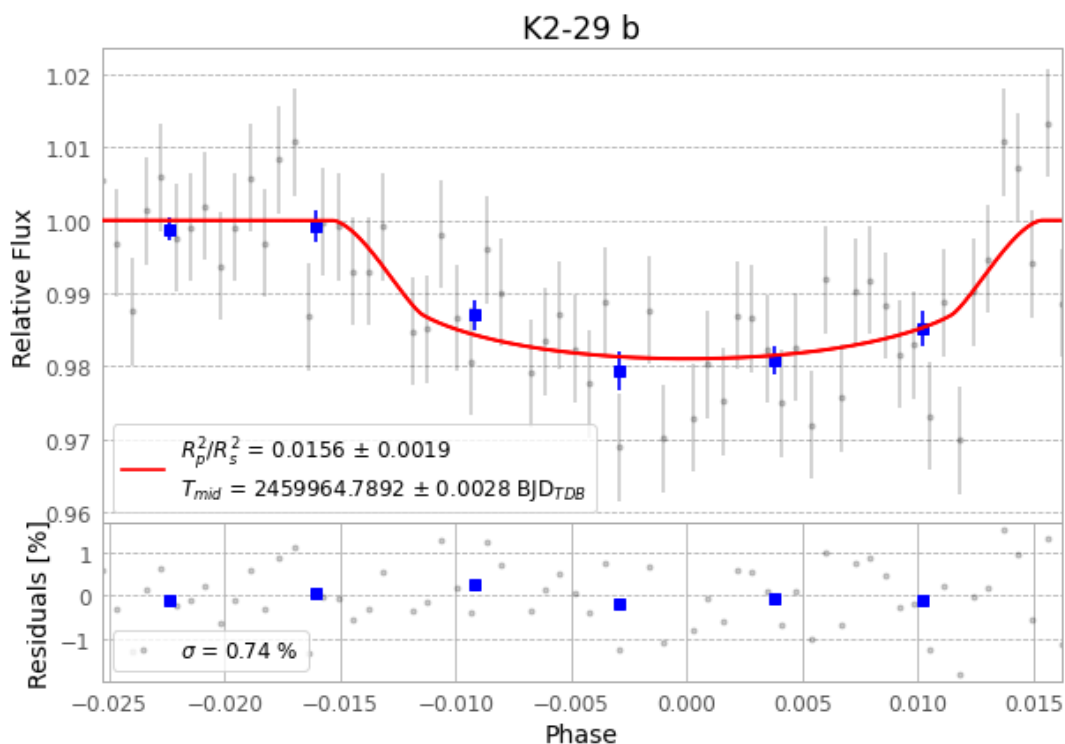


Figure 1 — FinalLightCurve_K2-29 b_2023-01-19.png (SHA-256 bf94721bb608cef8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 281], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 193349942bf38eaa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), K2-29230120045115.FITS → K2-29230120080615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-230120.log (847639f3a19d...), FinalLightCurve_K2-29 b_2023-01-19.png (bf94721bb608...), FinalLightCurve_K2-29 b_2023-01-19.csv (dcffe8f808ba...)

Compute resources

Wall time 171.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:38Z.